

U.S. Customary System to Metric System

1 inch (in.)	=	25.4 millimeters (mm)
1 inch (in.)	=	2.54 centimeters (cm)
1 foot (ft)	=	30.48 centimeters (cm)
1 foot (ft)	=	0.305 meters (m)
1 yard (yd)	=	91.44 centimeters (cm)
1 yard (yd)	=	0.914 meters (m)
1 mile (mi)	=	1609.34 meters (m)
1 mile (mi)	=	1.609 kilometers (km)

Use the conversion rates to convert between the lengths below.

1. 27 yards = 2468.88 centimeters

$$\frac{27 \text{ yd}}{1} \left| \frac{91.44 \text{ cm}}{1 \text{ yd}} \right| = 2468.88$$

2. 18 miles = 28.962 kilometers

$$\frac{18 \text{ mi}}{1} \left| \frac{1.609 \text{ km}}{1 \text{ mi}} \right| = 28.962$$

3. 58 feet = 17.69 meters

$$\frac{58 \text{ ft}}{1} \left| \frac{0.305 \text{ m}}{1 \text{ ft}} \right| = 17.69$$

4. 34 inches = 863.6 millimeters

$$\frac{34 \text{ in}}{1} \left| \frac{25.4 \text{ mm}}{1 \text{ in}} \right| = 863.6$$

5. If 1 yard = 91.44 centimeters, then 8.4 yards = 768.096 centimeters.

$$\frac{1}{91.44} = \frac{x}{768.096}$$

$$\frac{768.096}{91.44} = \frac{91.44x}{91.44}$$

$$8.4 = x$$

Metric System to U.S. Customary System

1 millimeter (mm)	=	0.039 inches (in.)
1 centimeter (cm)	=	0.394 inches (in.)
1 centimeter (cm)	=	0.033 feet (ft)
1 meter (m)	=	39.37 inches (in.)
1 meter (m)	=	3.281 feet (ft)
1 meter (m)	=	1.094 yards (yd)
1 kilometer (km)	=	1093.613 yards (yd)
1 kilometer (km)	=	0.621 miles (mi)

6. 540 meters = 590.76 yards

$$\frac{540 \text{ m}}{1} \left| \frac{1.094 \text{ yd}}{1 \text{ m}} \right| = 590.76$$

7. 91 kilometers = 56.511 miles

$$\frac{91 \text{ km}}{1} \left| \frac{0.621 \text{ mi}}{1 \text{ km}} \right| = 56.511$$

8. 25 centimeters = 0.825 feet

$$\frac{25 \text{ cm}}{1} \left| \frac{0.033 \text{ ft}}{1 \text{ cm}} \right| = 0.825$$

9. 800 millimeters = 31.2 inches

$$\frac{800 \text{ mm}}{1} \left| \frac{0.039 \text{ in}}{1 \text{ mm}} \right| = 31.2$$

10. If 1 kilometer = 1,093.613 yards, then 7,655.291 yards = 7 kilometers.

$$\frac{1}{1093.613} = \frac{x}{7655.291} \quad \frac{7655.291}{1093.613} = \frac{\cancel{1093.613}x}{\cancel{1093.613}} \quad 7 = x$$